

Function (Solve all the following problems using function):

1. Write a C program to find power x of an integer N (N^x), where N, x are integer numbers ($N, x \in \mathbb{Z}$).
2. Write a C program to find the maximum among three numbers and write another function to find the minimum of these three numbers.
3. Write a C program that has a function to find the maximum between two numbers and **use that function** to find the maximum among three numbers.
4. Write a C program to swap and print the values of two variables.
5. Write a C program to swap and print the values of two variables (without using any extra variables).
6. Write a C program to find the N^{th} prime number.
7. Write a C program to find all prime numbers between two numbers **using a function that returns 1 if the given input integer is a prime number, 0 otherwise**.
8. Write a C program to calculate the sum of n numbers coming from the console.
9. Write a C program to convert a number N in source base s to its equivalent number in destination base d ($1 < s, d < 11$).
10. Consider the following series:

$$1 + \frac{3!}{2^2} + \frac{5!}{3^3} + \frac{7!}{4^4} + \dots$$

Now take a number n as input. Output the n^{th} term of this series.

Recursive Function (Solve all the following problems using recursion)

11. Write a C program to print the numbers from 1 to N .
12. Write a C program to print the even, odd numbers (separately) in a given range $[M, N]$.
13. Write a C program to calculate the sum of numbers from 1 to N .
14. Write a C program to find power x of an integer N (N^x), where N, x are integer numbers ($N, x \in \mathbb{Z}$).
15. Take an integer N as input and find its factorial ($N!$).
16. Write a C program to count the number of digits of a number N .
17. Write a C program to print the reverse of a number N .
18. Write a C program to generate the reverse of a number N .

19. Write a C program to find the GCD of two numbers.
20. Write a C program to find the LCM of two numbers.
21. Write a C program that takes as input a positive integer n and outputs the number of proper divisors of n.
22. Write a C program to find the N^{th} fibonacci number. Assume that the fibonacci series starts with 0, 1, 1, 2, 3....
23. Write a C program to find the power of two present in a number.
24. Write a C program to count the number of zeros present in a number N.
25. Write a C program to remove the zeros present in a number N and generate the modified number.